# CHECKOUT GUARDIAN

Product Requirements Document (PRD) — Version 1.0

**Project:** Checkout Guardian

**Author:** Engineering & Product Team

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# 1. Executive Summary

**Product Overview:** Checkout Guardian is an advanced AI-powered buyer-protection and transaction-verification system designed to safeguard online consumers against sophisticated e-commerce fraud before payment completion.

**Problem Being Solved:** Modern cybercriminals exploit consumer trust by deploying professional-grade fake e-commerce storefronts, executing brand impersonation attacks, advertising unrealistic discounts, and deceptively routing customer funds to unauthorized third-party payment recipients.

**Proposed Solution:** Checkout Guardian acts as an intelligent pre-payment gatekeeper. By analyzing website authenticity, domain age, merchant credentials, advertised product pricing, and payment-

recipient consistency in real time, the system intervenes prior to payment capture.

**Target Users:** Everyday online shoppers, bargain hunters navigating unfamiliar digital storefronts, and consumers purchasing high-ticket items across global e-commerce channels.

**Core Value Proposition:** Real-time risk transparency and pre-payment transaction intervention that prevents fraudulent fund transfers before financial loss occurs.

**Key Differentiator:** The proprietary *Merchant Identity & Offer Consistency Engine*, which cross-references the claimed storefront brand against the actual underlying payment recipient.

## 2. Problem Definition

E-commerce fraud has evolved beyond rudimentary phishing pages into pixel-perfect storefront clones. Checkout Guardian addresses specific systemic vulnerabilities in the digital purchasing lifecycle:

- **Fake E-Commerce Websites:** Rapidly spun-up rogue web domains masquerading as legitimate retail outlets.
- **Brand Impersonation:** Unauthorized use of registered trademarks, stolen product imagery, and forged trust seals.
- **Misleading Discounts:** Predatory pricing models designed to induce impulsive checkout behaviors.
- **Unverified Merchants:** Absence of verifiable corporate registration or legal entity backing.
- **Seller Identity Mismatches:** Discrepancies between the displayed merchant entity and the legal beneficiary of funds.
- **Payment Recipient Inconsistencies:** Routing payments to unverified personal accounts or shell company accounts rather than the legitimate merchant entity.

> **Core Insight:** Existing checkout experiences validate payment instrument validity (e.g., card balance or PayPal account status) but fail to verify whether the underlying merchant is legitimate or if the funds are being directed to the entity the customer intends to pay.

## 3. Product Vision and Objectives

**Long-Term Vision:** To establish a universal trust and verification standard across global digital payments and checkout workflows.

**Hackathon Objectives:** Deliver a functional, high-impact MVP demonstrating real-time merchant verification, AI-driven risk scoring, and payment capture control using PayPal Sandbox integration.

| Objective Category | Target Metric / Milestone |
|---|---|
| Primary Goal | Successfully intercept and pause payment capture on simulated fraudulent merchant profiles. |
| Secondary Goal | Provide transparent, explainable risk rationale to users within <2 seconds. |
| Success Indicator | 100% accurate classification across predefined demo test scenarios (Genuine, Suspicious, Fraudulent). |

## 4. Target Users and Personas

- **Persona 1: The Bargain Hunter (Regular Online Shopper)** — Frequently discovers niche web stores via social media ads. Vulnerable to deep discounts on unverified sites. Needs automated protection without friction during genuine transactions.
- **Persona 2: The High-Ticket Purchaser** — Acquiring electronics, designer goods, or appliances from unfamiliar merchants. Highly concerned with financial security, warranty validity, and recourse in case of fraud.
- **Persona 3: The Vulnerable Digital Consumer** — Less technically adept at inspecting SSL certificates, WHOIS registration records, or hidden business entity registrations. Relies on clear visual risk warnings ("Allow", "Warn", "Block").
- **Persona 4: Payment Ecosystem Stakeholders (Merchant / PayPal Ecosystem)** — Interested in reducing chargeback fraud, preserving platform trust, and mitigating reputational damage caused by malicious actors abusing payment gateways.

## 5. Proposed Solution and Core Features

Checkout Guardian comprises ten modular product features, anchored by the central innovation:

### 1. Website Authenticity Verification

Inspects Document Object Model (DOM) elements, favicon integrity, contact page validity, and legal disclosure presence.

## 2. Domain and Reputation Analysis

Queries domain registration age, registrar reputation history, and SSL certificate issuer metadata.

## 3. Merchant Identity Verification

Cross-references business registration claims against trusted corporate registries and digital footprint databases.

## 4. Product and Price Intelligence

Evaluates historical product pricing norms to flag predatory, outlier discounts designed to manufacture false urgency.

## 5. Merchant Identity & Offer Consistency Engine (Central Innovation)

Compares the brand identity presented on the storefront against the actual merchant ID and payee entity returned by the payment gateway API during checkout.

## 6. AI-Based Risk Assessment

Combines deterministic heuristic checks with LLM-powered contextual reasoning to synthesize multi-signal risk profiles.

## 7. Explainable Risk Reports

Generates clear, natural-language explanations detailing why a specific website or transaction was flagged.

## 8. Pre-Payment Warning and Intervention

Pauses payment capture workflows and presents an interactive safety overlay when high-risk anomalies are detected.

## 9. Customer Protection Dashboard

Provides users with historical audit logs, past risk scans, and security status summaries.

## 10. Browser Integration / Demo Application Portal

Seamlessly integrates into the checkout flow to monitor transaction payloads and enforce safety controls.

## 6. End-to-End User Journey

1. **Discovery & Navigation:** Customer browses an online shopping portal via the integrated shopping environment or browser extension.
2. **Automated Ingestion:** Checkout Guardian extracts page metadata, domain details, and product pricing information.
3. **Preliminary Evaluation:** The system conducts domain age lookup, SSL verification, and heuristic checks.
4. **Checkout Initiation:** Customer selects PayPal as their payment method and proceeds to checkout.
5. **Recipient Extraction:** Integration intercepts order initialization payloads to identify the actual payment recipient (Merchant ID/Account).
6. **Consistency Matching:** The *Merchant Identity & Offer Consistency Engine* compares the displayed store brand with the payment payee.
7. **Decision Engine Execution:** Multi-signal AI assessment determines risk tier (Low, Medium, High).
8. **Intervention & Resolution:**
   - *Genuine Store:* Transaction proceeds smoothly.
   - *Suspicious Offer:* Warning banner displayed with optional user override.
   - *Fraudulent Store / Mismatch:* Payment capture paused immediately; user protected from financial loss.

## 7. Functional Requirements

| ID | Requirement Description | Priority | Acceptance Criteria |
|---|---|---|---|
| FR-001 | Website Information Collection | Must Have | Extracts URL, title, meta descriptions, and contact info within 500ms of page load. |
| FR-002 | Domain Authenticity Analysis | Must Have | Retrieves domain creation date and registrar information via lookup API. |

| | | | |
|---|---|---|---|
| FR-003 | Merchant Identity Verification | Must Have | Validates claimed merchant details against structured entity records. |
| FR-004 | Product and Price Analysis | Should Have | Flags price deviations exceeding 70% below baseline market averages. |
| FR-005 | Payment Recipient Verification | Must Have | Compares checkout payee ID with claimed merchant entity. |
| FR-006 | AI Risk Assessment | Must Have | Synthesizes signals into Low, Medium, or High risk classifications. |
| FR-007 | Risk Explanation Generation | Must Have | Outputs structured, human-readable rationale for risk scores. |
| FR-008 | Payment Intervention | Must Have | Successfully pauses payment capture in PayPal Sandbox upon high-risk triggers. |
| FR-009 | Customer Dashboard | Could Have | Displays historical scan results and active protection metrics. |
| FR-010 | Transaction & Risk Logging | Must Have | Stores encrypted audit logs of risk evaluations and user actions. |

## 8. Non-Functional Requirements

- **Performance & Latency:** Risk assessment and domain analysis complete in < 2.0 seconds to prevent checkout friction.
- **Security:** All API communications encrypted via TLS 1.3; zero raw storage of payment credentials or sensitive personal identifiable information (PII).
- **Reliability & Availability:** 99.9% uptime target for backend evaluation services during hackathon demonstration.
- **Explainability:** Risk determinations must provide transparent, itemized evidence rather than opaque scores.
- **Error Handling:** Graceful degradation fallback if third-party domain lookup or LLM APIs experience transient timeouts.

## 9. AI and Risk Assessment Requirements

Checkout Guardian employs a hybrid architecture combining deterministic verification rules (WHOIS lookups, SSL certificate checks, exact payee matching) with LLM-powered semantic analysis (evaluating promotional language, anomaly detection in product descriptions, and generating natural language risk explanations).

> **Uncertainty Handling:** LLM outputs are treated as probabilistic indicators rather than definitive proof of fraud. The decision engine requires corroborating deterministic evidence before triggering high-risk payment pauses.

## 10. PayPal Integration Requirements

PayPal serves as the foundational payment rail for Checkout Guardian:

- **Integration Stack:** PayPal JavaScript SDK, Orders API, and Sandbox environment.
- **Workflow Control:** The backend validates order payloads prior to authorizing final payment capture.
- **Capture Interception:** In high-risk scenarios, backend logic bypasses capture execution and triggers a security intervention prompt.
- **Scope Clarification:** The hackathon demonstration controls capture within the integrated application environment, demonstrating architectural feasibility for future gateway-level integration.

## 11. System Components & Architecture Overview

| Component Layer | Technology Stack | Core Responsibility |
|---|---|---|
| Frontend | React.js / Chrome Extension | User interface, dashboard, and checkout overlay injection. |
| Backend API | Node.js & Express.js | Orchestration, routing, and request validation. |

| | | |
|---|---|---|
| Risk Engine | Python / Node.js & LLM API | Heuristic aggregation and risk scoring logic. |
| Database | PostgreSQL / MongoDB | Audit logging, session state, and domain cache. |
| Payment Gateway | PayPal Sandbox SDK / APIs | Order processing and payment capture management. |

## 12. Risk Scoring and Decision Framework

The risk framework evaluates multiple weighted vectors:

- **Domain Age & Reputation (30%):** Newly registered domains (< 30 days) receive elevated risk weight.
- **Merchant Identity Consistency (40%):** Mismatches between claimed storefront and payment recipient trigger immediate high-risk flags.
- **Pricing Anomalies (20%):** Unrealistic discounts (>80% off) increase suspicion score.
- **SSL & Technical Indicators (10%):** Missing or mismatched SSL certificates.

**Decision Thresholds (Subject to calibration):** Score 0–39 (Low Risk: Allow), Score 40–74 (Medium Risk: Warn), Score 75–100 (High Risk: Pause Capture).

## 13. Hackathon MVP Scope

- **Must Have:** Website metadata ingestion, PayPal Sandbox checkout flow, Merchant Identity & Offer Consistency check, risk scoring engine, and payment pause intervention.
- **Should Have:** Explainable risk report generation and clean dashboard UI.
- **Could Have:** Advanced browser extension pop-up widget.
- **Out of Scope:** Universal gateway interception across arbitrary external websites without SDK integration.

## 14. Demo Scenarios & Expected Results

| Scenario | Input Profile | Expected Result |
|---|---|---|
| **Scenario A: Genuine Store** | Established domain, verified merchant, matching PayPal payee. | **Allow Transaction** |
| **Scenario B: Suspicious Offer** | Recent domain, deep discount, unverified minor seller. | **Warning & User Prompt** |
| **Scenario C: Fake Impersonator** | Brand clone domain, mismatched payment recipient entity. | **High Risk — Pause Capture** |

## 15. Success Metrics and Evaluation

- 100% successful detection and capture-pause on designated fraudulent demo test cases.
- End-to-end risk evaluation latency under 2.0 seconds.
- Zero false positives on benchmark genuine merchant test cases.

## 16. Privacy, Security and Ethical Considerations

Strict data minimization principles are enforced. No financial account passwords or raw card numbers are stored or processed by Checkout Guardian. All API communications operate over secure HTTPS/TLS channels. User warnings are designed to be objective, transparent, and non-defamatory.

## 17. Risks, Dependencies and Limitations

External dependencies include WHOIS lookup reliability, third-party LLM API stability, and PayPal Sandbox uptime. The prototype focuses on controlled merchant verification and does not claim universal protection across unrestricted web traffic.

## 18. Development Roadmap (Hackathon Timeline)

- **Phase 1:** Architecture setup, repository initialization, and database schema design.
- **Phase 2:** Core domain analysis and Merchant Identity & Offer Consistency Engine development.

- **Phase 3:** PayPal Sandbox integration and payment capture control hook implementation.
- **Phase 4:** Frontend UI development, risk dashboard, and warning modal integration.
- **Phase 5:** End-to-end testing, scenario validation, and demo preparation.

## 19. Acceptance Criteria & Definition of Done

The project is deemed complete for the hackathon when all Must-Have functional requirements are demonstrated live across Scenarios A, B, and C, with verifiable payment capture pausing in PayPal Sandbox and clear, explainable risk outputs displayed to the user.

## 20. Open Questions and Assumptions

> **Key Assumption:** Sandbox API merchant identifiers accurately reflect real-world merchant metadata structures for demonstration purposes. Future commercial iterations will require direct integration with global corporate registry verification services.

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